CSD LAB 2

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Assigned system: stu80_sys1

Flask source code:

```
from flask import Flask, request, Response
import cv2
import numpy as np

app = Flask(__name__)

def read_image():
    file = request.files["image"]
    img = cv2.imdecode(
        np.frombuffer(file.read(), np.uint8),
        cv2.IMREAD_COLOR
    )
    return img


@app.post("/gray")
def gray():

    img = read_image()

    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    _, out = cv2.imencode(".jpg", gray)

    return Response(out.tobytes(),
                                    mimetype="image/jpeg")


@app.post("/blur")
def blur():

    img = read_image()

    blur = cv2.GaussianBlur(img, (11,11), 0)

    _, out = cv2.imencode(".jpg", blur)
```

```
        return Response(out.tobytes(),
                                        mimetype="image/jpeg")


@app.post("/edges")
def edges():

        img = read_image()

        gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

        edge = cv2.Canny(gray,100,200)

        _, out = cv2.imencode(".jpg", edge)

        return Response(out.tobytes(),
                                        mimetype="image/jpeg")


@app.post("/contours")
def contours():

        img = read_image()

        gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

        edge = cv2.Canny(gray,100,200)

        cnts, _ = cv2.findContours(
                edge,
                cv2.RETR_EXTERNAL,
                cv2.CHAIN_APPROX_SIMPLE
        )

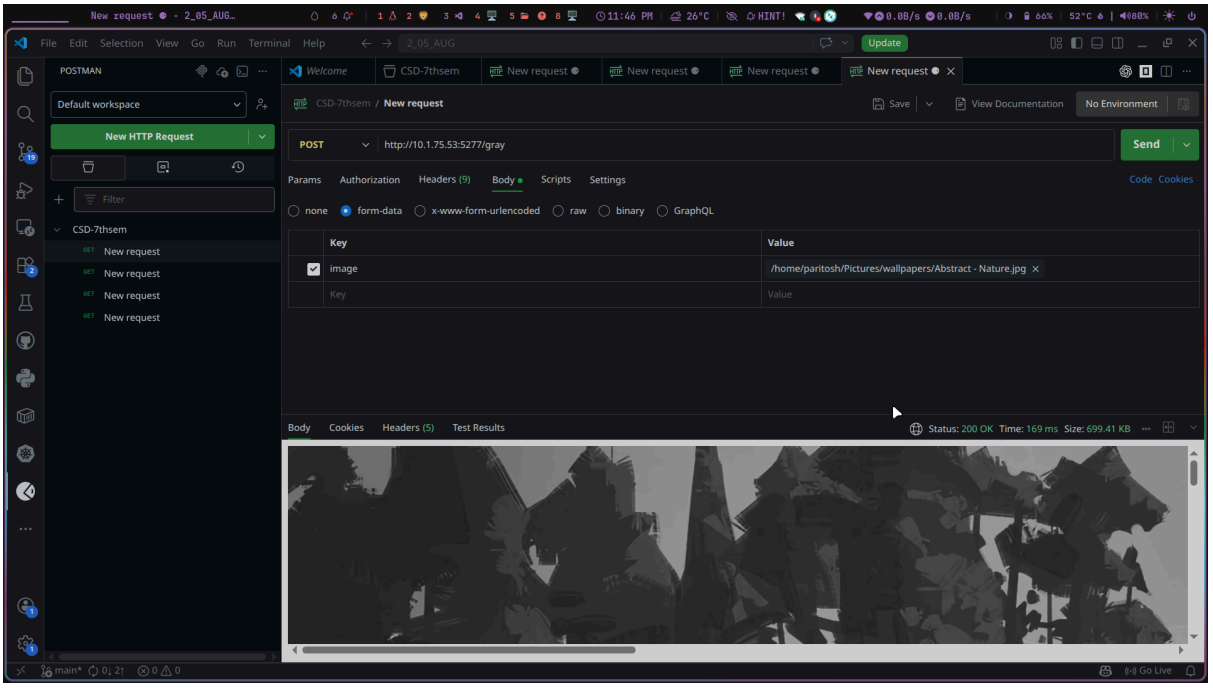
        cv2.drawContours(img, cnts, -1, (0,255,0), 2)

        _, out = cv2.imencode(".jpg", img)

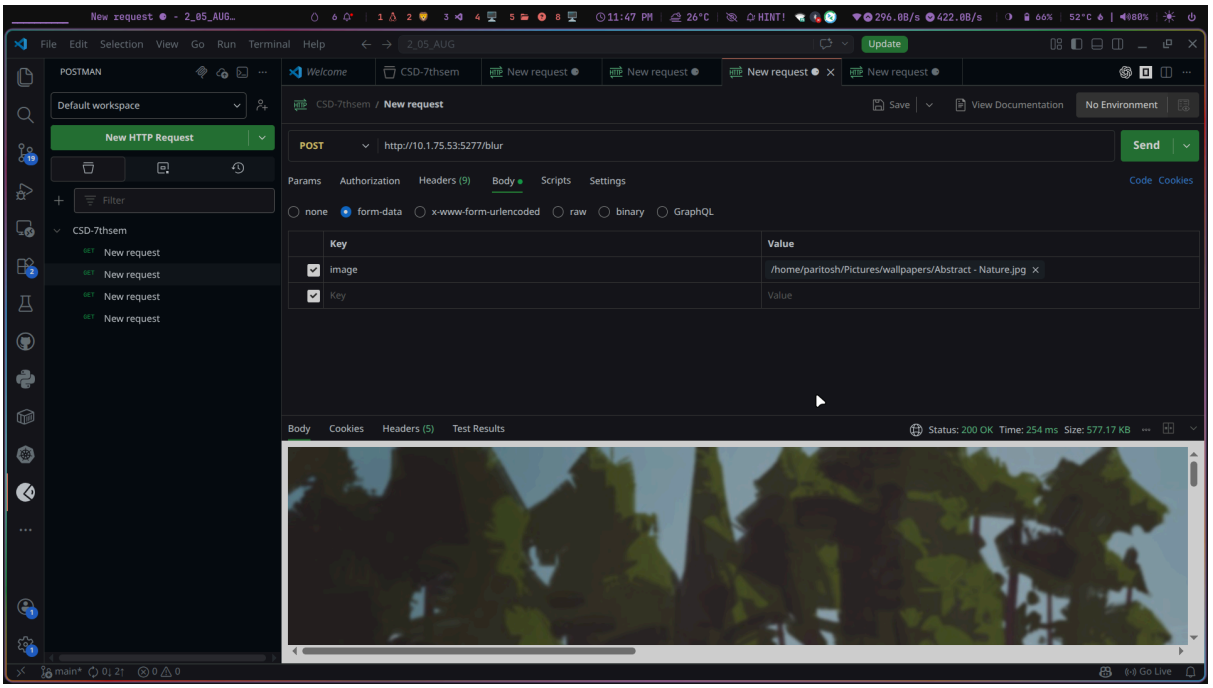
        return Response(out.tobytes(),
                                        mimetype="image/jpeg")


app.run(host="0.0.0.0", port=5000)
```

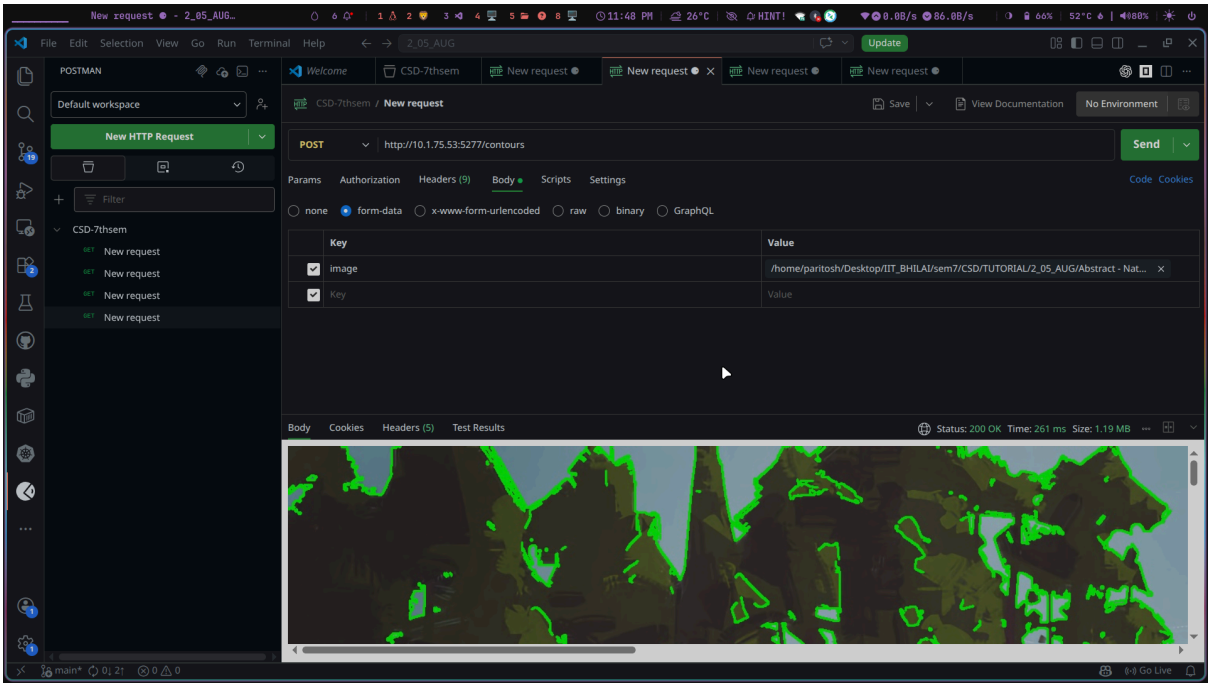
POSTMAN screenshots:



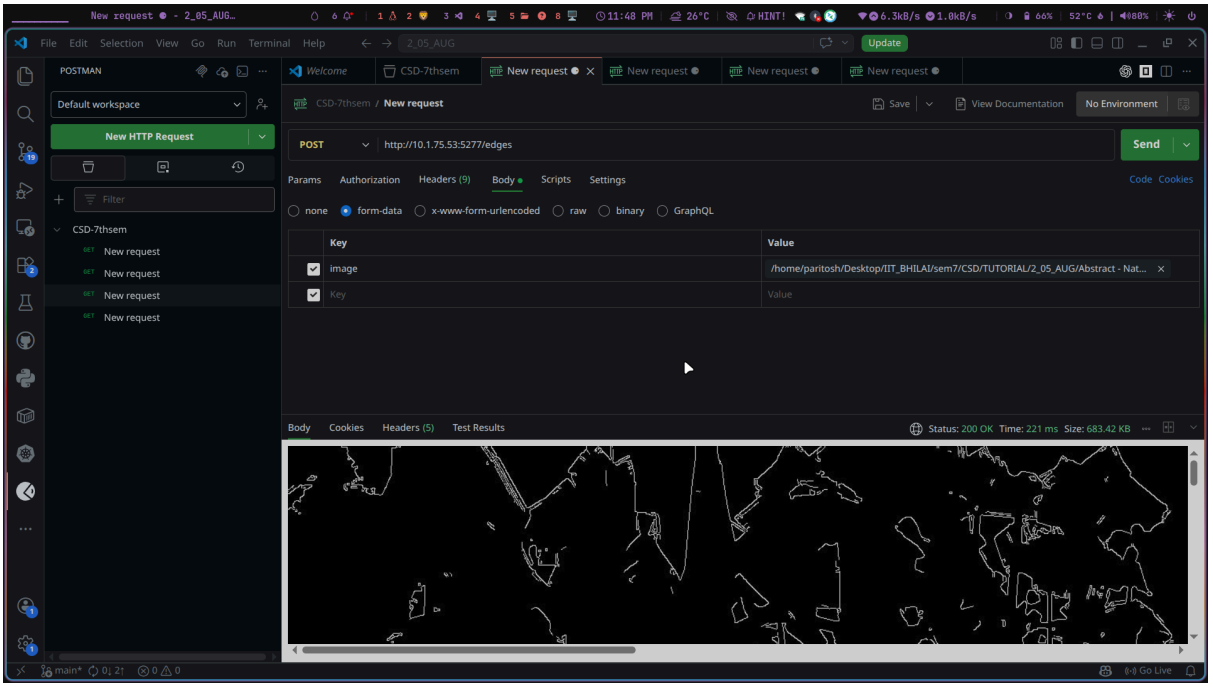
(/gray)



(/blur)



(/contours)



(/edges)

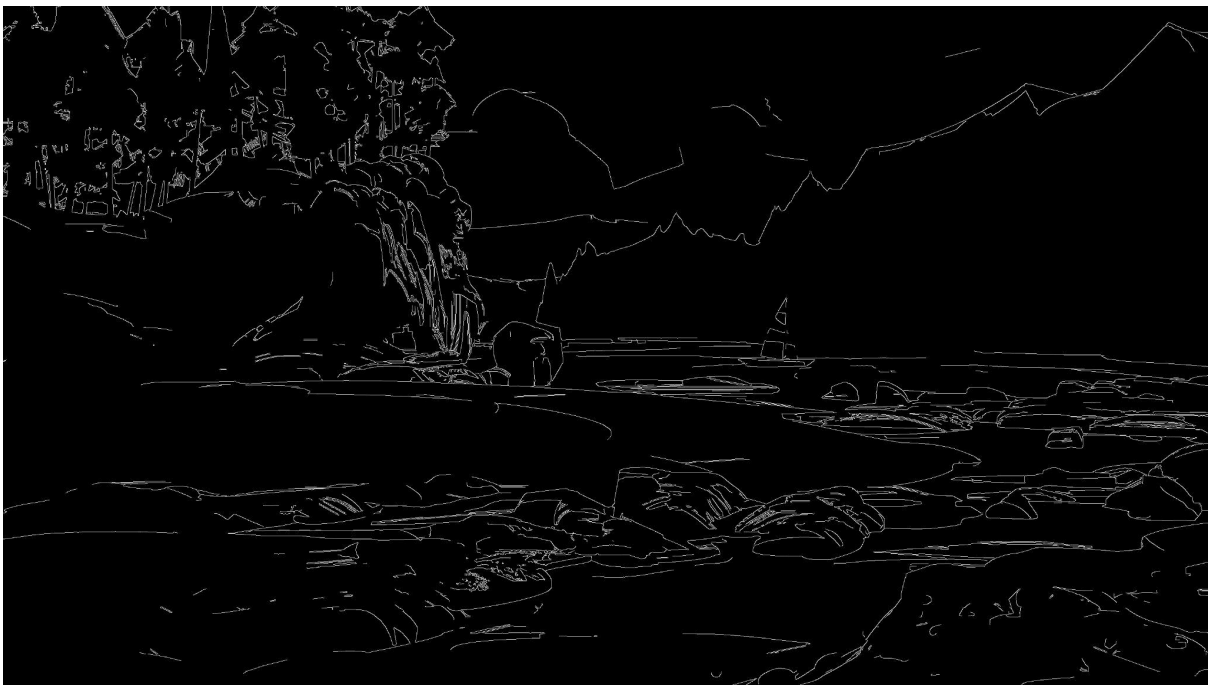
Input and output images:



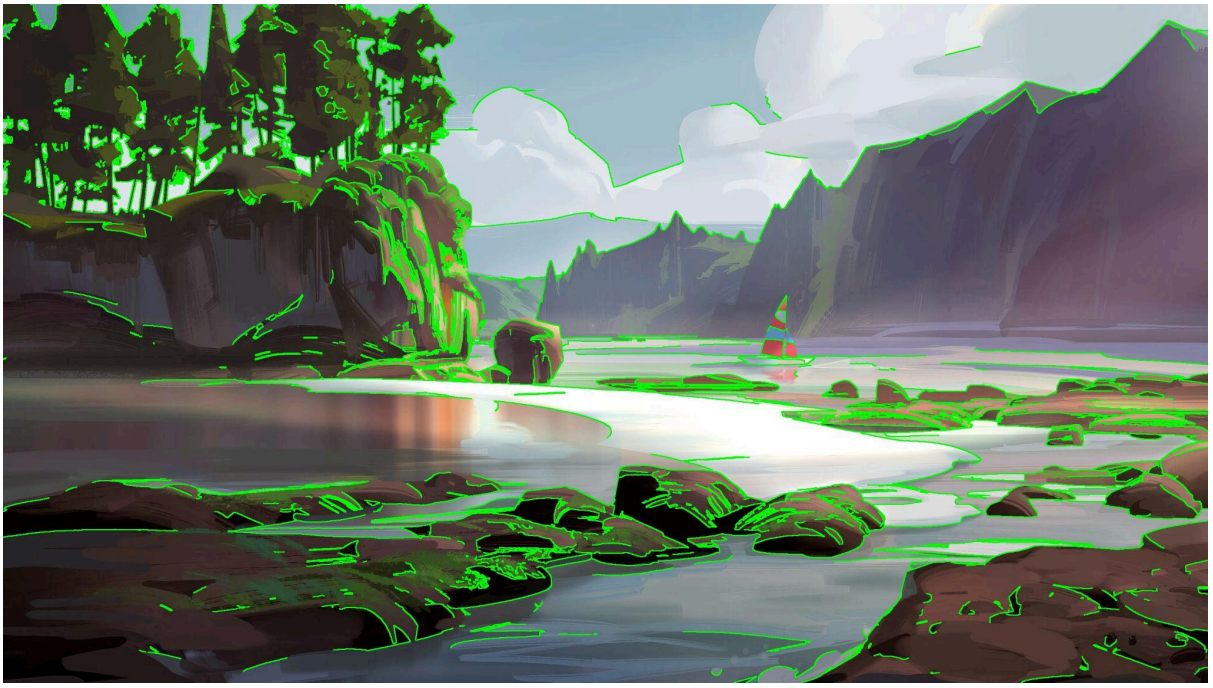
(INPUT)
(I have used the same image for all post request)



(OUTPUT - GRAY)



(OUTPUT - EDGES)



(OUTPUT - CONTOURS)



(OUTPUT - BLUR)

I have used only one port 2277 (4 POST requests)